

Máté Kaposi

Mechatronics Engineering Student

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Personal Information

Date of Birth

21 August 2001

Nationality

Hungarian

Technical Skills

Programming

C, C++, C++/CLI, Python, MATLAB, Simulink, LabVIEW, SQL

CAD & Engineering Software

Autodesk Inventor, Autodesk EAGLE, MicroStation, ANSYS Workbench, Confluence, Jira, Git, Linux

Image Processing

OpenCV, MeshLab, Fiji

Languages

Hungarian

Native

English

B2 – Internationally recognised language certificate

German

B2 – Internationally recognised language certificate

Additional Information

Driving Licence

Category B

Competition

2023 – Plan@Tech AUDI Engineering Competition

Best Team Result and Best Individual Result

Selected Engineering Projects

Spring-Disk Mechanical System

LabVIEW

Developed a mathematical and simulation model of a spring-disk mechanical system and analysed its vibrational behaviour using LabVIEW.

Microcontroller-Based Temperature Measurement System

C, C++/CLI, EAGLE

Designed a microcontroller circuit and implemented the communication protocol required for the Dallas HW-477

Education

2026

MSc in Mechatronics Engineering

Saitama University, Japan

Exchange Program



2025 – Present

MSc in Mechatronics Engineering

Budapest University of Technology and Economics (BME)

Specialisation: Embedded Systems



2021 – 2025

BSc in Mechatronics Engineering

Budapest University of Technology and Economics (BME)

Specialisation: Biomechanics



Work Experience

July 2025 – April 2026

Project Manager Intern

aiMotive



- Developed and optimized SQL queries for database management and reporting.
- Collaborated with the Calibration Team to implement small features in C++.
- Developed Python scripts for data management, test scripts, and test data.
- Designed Python-based tools to calculate and evaluate road network coverage metrics across international regions.

June 2024 – January 2025

Manufacturing Design Intern

Audi Hungaria Zrt.



temperature sensor. Developed a graphical interface for data visualisation and analysis.

Characterisation of a Frictional Oscillator

MATLAB

Investigated the sticking behaviour of a frictional oscillator using one-dimensional mappings and reconstructed system stability from the identified sticking points.

Contactless Respiratory Tracking System Python, OpenCV

Designed an intensity-based contactless respiratory tracking system for a CyberKnife radiation therapy system. Developed image-processing and tracking methods using Python and OpenCV.

- Designed transport carts for the PPE engine production line.
- Prepared technical specifications for procurement and coordinated communication with suppliers.
- Conducted FMEA analyses for the designed transport carts.
- Updated existing technical drawings in MicroStation with the new cart positions and related design information.

Profile

Mechatronics engineering student with a strong interest in embedded systems, robotics, automation, image processing, and software engineering. Experienced in mechanical design, programming, simulation, and experimental engineering projects. Interested in applying engineering knowledge to practical technological challenges.